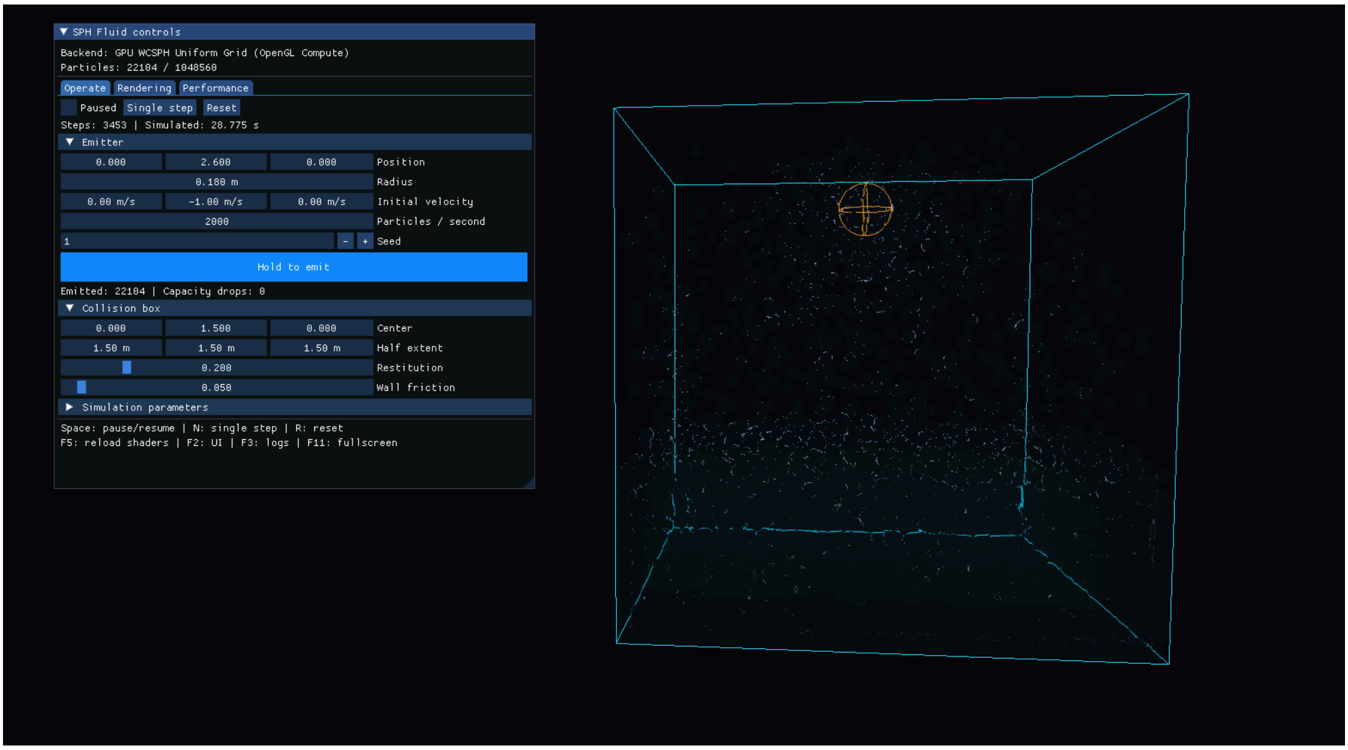
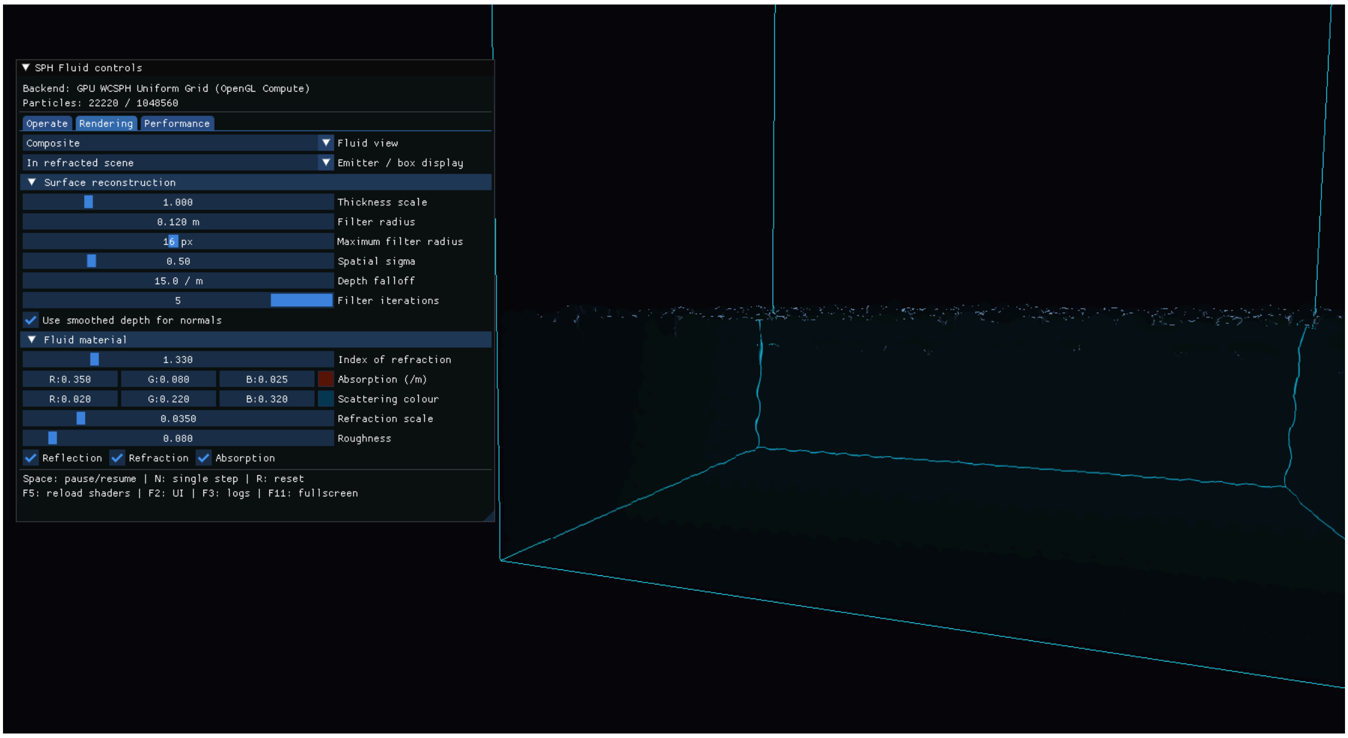
Real-Time SPH Fluid Simulation





Project Goal

This project implements an interactive WCSPH fluid simulation in C++17 and OpenGL. It includes CPU and GPU solvers, uniform-grid neighbour search, multithreaded CPU simulation, and screen-space fluid rendering.

Code

- [Project source]()

- [Simulation code]
- [Rendering code]
- [Compute shaders]
- [Tests and benchmark results]

The project uses CG_Labs for window creation, input, camera control, ImGui, shader management, and CMake dependencies.

Build

This project is built as part of CG_Labs. Follow the repository setup and Visual Studio instructions in:

- [CG_Labs README]
- [CG_Labs build guide]

Usage

1. Open the CG_Labs repository folder in Visual Studio and wait for **CMake generation finished.**
2. In the target dropdown next to the green Run button, select `src\SPHFluid\SPH_Fluid.exe`. Do not select an `(Install)` target.
3. Press `F5` to build and run the project.

The ImGui interface can be used to control the particle emitter, collision box, SPH parameters, rendering mode, and performance display.

Input	Action
`Space`	Pause or resume the simulation
`N`	Advance one simulation step
`R`	Reset the simulation
`F2`	Show or hide the controls
`F5`	Reload shaders